



Hardware Sustainability

2022-05-26

Student materials

- web
- PDF

Overview

Ethics background required: Ideally, students should be familiar with the 4 ethical frameworks presented in the first year curriculum (virtue ethics, deontology, consequentialism, analogies). If they (or the professor) are not, a brief summary is provided in the reflection.

Subject matter referred to in this lab: technological devices and sustainability.

Placement in overall ethics curriculum:

- This lab is suitable for any course considering the designing of a product and the impact on the environment. We recommend a 3rd or 4th year computer architecture or systems course or similar engineering course.
- Recommended previous labs: Foundational labs.
- Recommended follow-up labs: None

Time required:

- Out of class: None;
- In class: 45 minutes

Learning objectives: Students will

- Recognize ethical issues and dilemmas that surround prolific production of technology.
- Apply different frameworks to the dilemma of whether to continue the production of technology at the speed with which we have been.
- Articulate independent conclusions regarding the ethical dilemma.

Ethical dilemma or issue to be considered:

- Dilemma: Should we continue to produce technology at the current rate in the U.S.?
- Issue: sustainability, e-waste.



With more technological innovation, people are able to do things they never could before. Healthcare is available to rural areas. Businesses are making more money. However, as humans create more technology, we also create more e-waste. Should we continue to develop new technology knowing it creates more waste? Where and how should we dispose of our e-waste? This lab is designed to get students to identify one or more ethical dilemmas and evaluate them through an ethical framework.

Flow

- Identification of possible issues: Students will place their electronics on their desk in front of them as a visual of the amount of technology used by a single person.
- Students and professor will explore the value and challenges of the prolific production of electronic devices.
- Students will reflect on the ethicality of such production and resulting disposal requirements using a variety of ethical frameworks.

Preparation: Read through the entire lab. Prepare a slide for the introduction (if needed) and print out student handouts.

Guide for Instructors

Lesson plan

Introduction (5 minutes)

Read or summarize to students: Today we are going to look at the ways that the prolific production of new technology affects our lives. New devices increase our ability to communicate, provide healthcare to rural areas, and make our lives more fun and efficient. However, there are downsides as well. Some of those downsides are ethically questionable. We will consider together these ups and downs and questions to consider when new technology is produced.

- Have students place their phones, laptops, calculators, chargers, and any other electronics all together on their desk or table in front of them; allow students to observe and note how much electronics exist in just one class.

Activity (20 minutes)

Divide the class into groups of 2-3 and assign each group to answer one of the 7 questions on the student handout by looking for resources online. We have provided some answers that were correct when the lab was written just to give you, the instructor, an idea. (10 minutes)

Question #1: How have personal electronic devices improved medicine?

- Electrocardiogram on your phone, send blood glucose levels via phone



<https://www.carecloud.com/continuum/7-best-fda-approved-health-apps/>

- Detecting skin cancer and ear infections

<https://www.nbcnews.com/mach/science/smartphones-are-changing-medical-care-some-surprising-ways-ncna810561>

Question #2: How have electronics in general improved medicine?

- Smart inhalers, robotic surgery, artificial organs (using 3D printers)

<https://www.proclinical.com/blogs/2019-2/top-10-new-medical-technologies-of-2019>

Question #3: How have electronic devices improved the environment?

- Carbon collection devices controlled by AI

<https://www.forbes.com/sites/bernardmarr/2019/02/27/bill-gates-reveals-the-10-breakthrough-technologies-that-will-change-the-world-in-2019/#6b3c1002171d>

- Ability to find and manage finite natural resources

<https://www.forbes.com/sites/kinetica/2019/05/31/the-internet-of-things-is-powering-the-data-driven-fourth-industrial-revolution/#3fbec2c24e8>

Question #4: How many mobile devices do individuals have on average? Why are people not keeping their devices longer?

- More mobile devices than people

<https://www.bankmycell.com/blog/how-many-phones-are-in-the-world>

- People want the new features, and sometimes phones are required to be replaced rather than repaired

<https://www.ewaste1.com/benefits-of-e-waste-recycling/>

Question #5: What is e-waste, and how much e-waste accumulates in one year?

- 176 pounds per family per year.

<https://news.nationalgeographic.com/2017/12/e-waste-monitor-report-glut/>

Question #6: Can e-waste be recycled? What are the benefits?

- E-waste can and should be recycled. If left in landfills, the metals from the technology can contaminate the groundwater among other things

<https://www.ewaste1.com/benefits-of-e-waste-recycling/>

Question #7: Where is e-waste from the U.S. currently discarded?

- The U.S. currently disposes of its e-waste in other countries.



<https://www.pbs.org/newshour/science/america-e-waste-gps-tracker-tells-all-earthfix>

- Images may be found from the following sources (or feel free to use other reliable photos that you find)
 - <http://www.peter-nicholls.com/ghana-e-waste/u09c4nvsftyecx8y0y7svtfaxipwa8>
 - <https://www.aljazeera.com/indepth/inpictures/2014/01/pictures-ghana-e-waste-mecca-2014130104740975223.html>
 - <https://news.nationalgeographic.com/news/2014/06/140628-electronics-waste-india-pictures-recycling-environment-world/>
- As seen in some of the photos, people may mine the gold in discarded circuit boards
 - <https://www.statista.com/chart/12332/value-and-volume-of-raw-materials-in-e-waste-worldwide/>
 - <https://tokyo2020.org/en/games/medals/project/>
- Or they may repurpose the old electronics

Bring the students back together to report their answers to the assigned question (10 minutes).

Reflection as a class (20 minutes)

- Ask students what the ethical dilemma is.
 - They should see that discontinuing the creation of these devices could keep people from getting the help they need or making life less efficient. It might also eliminate jobs for those designing and manufacturing them. But, continuing the creation of these devices can be detrimental to the environment, particularly in other countries where the waste is sent. There are other problems as well: device addiction, people spending money they don't have etc.
- Question to the class: Assume that you are a high-level employee in a company that is considering creating a new version of an existing device. You are asked your opinion on whether this next version should go forward. Argue, using one of the ethical frameworks with which you are familiar (and are summarized on your handout) or the analogies tool, whether you should or should not move forward with this product. Feel free to state assumptions that you have made or questions that you would have for the supervisor asking for your input. Give 5 minutes to formulate their ideas using one framework.
- After the class has reconvened, go through each framework and get arguments from each person who chose that approach. Here are some possible ideas in case no one has one for a particular approach:
 - Virtue ethics: It would really depend on what this new device could contribute. If it would add some new accessibility feature for a marginalized portion of society, then



compassion, kindness, and inclusivity would be a focus. The virtuous person would probably also ask if a software upgrade could do the same thing to address the sustainability issue. If it was just additional entertainment value, or ease of use, the virtuous person might choose against the creation for sustainability reasons. At the same time they should think of a solution for keeping designers and manufacturers employed.

- Deontology: A set of rules might require a certain number of new features before a new device is actually made. It might require that all updates that can be included via software, be included. It might limit the number of hardware changes allowed each year. It might require manufacturers to use certain more easily recyclable materials if a new device was to be created. A student could also refer to a set of rules in existence. For example, the ACM code of ethics has principles “Do no harm” and “Give comprehensive and thorough evaluations of computer systems and their impacts, including analysis of possible risks.”
- Consequentialism: The stakeholders would be designers, manufacturers, consumers from various demographics, and the environment. Careful consideration would have to be paid to who the new features would benefit. Chances are that since a new device would benefit all employees and perhaps many others, the emphasis would need to be on reducing the cost – focusing on more sustainable materials, maybe at the cost of increased power, battery-life, or speed. Another way to reduce societal cost would be to find a lower population area to dispose of waste.

A student could, however, argue that the effects on the environment would harm not only people, but wildlife and greenery, thus making the costs the bigger impact

- Analogy might be cars and pollution.

Assessment

1. Identify the ethical dilemma concerning e-waste and technology.
2. Give two examples of how technology has benefited our society.
3. Current approaches to AI produce highly accurate results at the cost of executing an enormous amount of instructions thus contributing to a very high carbon footprint. Green AI is an area of research that tries to find a balance between accuracy and efficiency. Argue in support of Green AI using virtue ethics, deontology (is there a set of rules you can use?), or consequentialism.

See R. S. and J. Dodge, N. Smith, and O. Etzioni [1].

Online teaching

This lesson is easily adapted to an asynchronous online environment where a student would complete the assignment individually and then post some of their ideas while also responding to the ideas of others.



Inspired by work from D. Inclezan and L. Pradanos [2].

- Here are some other benefits to new technology
 - E-devices that track health
 - Better cameras
 - Stay connected
 - Don't get lost (GPS)
 - Natural Disaster Alerts
 - More entertainment
 - Voice commands aid people with disabilities
 - Opportunities to make money from home (eBay, Etsy)

References

Bibliography

- [1] R. S. and J. Dodge, N. Smith, and O. Etzioni, "Green AI," Communications of the ACM, Dec. 2020.
- [2] D. Inclezan and L. Pradanos, "Promoting Ecoliteracy in an Introductory Database Systems Course: Activities for the First Week," in Proceedings of the 45th ACM Technical Symposium on Computer Science Education, in SIGCSE '14. Atlanta, Georgia, USA: Association for Computing Machinery, 2014, pp. 573–578. doi: 10.1145/2538862.2538909.